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ORIGINAL RESEARCH ARTICLE

Premorbid Risk Factors and Acute Injury Characteristics of Sport-Related Concussion Across the National Collegiate Athletic Association: Findings from the Concussion Assessment, Research, and Education (CARE) Consortium

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Abstract

Background Previous sport-related concussion research highlights post-injury characteristics that influence recovery trajectories; however, there is limited information regarding premorbid factors that affect sport-related concussion risk.

Objective We aimed to (a) compare premorbid demographic factors among a large cohort of collegiate student athletes who did or did not sustain a sport-related concussion and (b) assess differences in acute injury characteristics based on biological sex and contact level.

Methods We conducted a cohort study of university student athletes from 22 sports enrolled in the Concussion Assessment, Research, and Education (CARE) Consortium study from 2014 to 2021 ($n = 1804$ student athletes with sport-related concussions; $n = 21{,}702$ student athletes without sport-related concussions).

Results Statistical analyses indicated student athletes who self-identified as Black (odds ratio [OR] = 1.61; 95% confidence interval [CI] = 1.42, 1.81) or multiracial (OR = 1.32; 95% CI 1.10, 1.59) demonstrated greater odds of experiencing sport-related concussions than White-identifying student athletes. Additional findings suggest male athletes (OR = 1.47; 95% CI 1.20, 1.81) and contact sport student athletes (OR = 1.40; 95% CI 1.16, 1.70) may be at increased odds for sport-related concussions if they were previously diagnosed with attention deficit hyperactive disorder. Notable post-injury characteristics across sexes included differences in the incident loss of consciousness (male: 5.9%, female: 2.6%; $p < 0.001$), post-traumatic amnesia (male: 13.6%, female: 5.1%; $p < 0.001$), and retrograde amnesia (male: 6.8%, female: 2.8%; $p < 0.001$). A greater proportion of contact-sport student athletes experienced an altered mental status (52.7%) than limited contact (36.2%) and non-contact (48.6%) [$p < 0.001$]. Last, student athletes participating at lower contact levels were more likely to have a longer delay in removal from activity following injury (contact: 73.6 ± 322.2 min; limited contact: 139.1 ± 560.0 min; non-contact: 461.4 ± 1870.8 min; $p = 0.005$).

Conclusions The present study provides contemporary pre- and post-sport-related concussion injury characteristics using a considerably sized cohort of collegiate student athletes. These findings support previous work suggesting sport-related concussion results in complex individualized clinical presentations, which may influence management strategies.

Members of the CARE Consortium Investigators are listed in "Acknowledgements" section.

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Key Points

Altered mental status including loss of consciousness, post-traumatic amnesia, and retrograde amnesia were once considered to be hallmark signs of sport-related concussions, yet there are limited data to determine whether these post-injury characteristics are as prevalent as previously estimated.

Biological sex, race, and medical histories such as previous sport-related concussions, learning disorders, attention deficit hyperactivity disorder, as well as migraine and non-migraine headache disorders may influence the risk for subsequent sport-related concussions.

Clinicians may utilize these findings by considering patient-specific intrinsic and extrinsic factors that may increase their risk for injury.

1 Introduction

Sport-related concussion (SRC) affects millions of athletes annually [1] with the greatest prevalence among those between the ages of 9 and 22 years owing to increased participation in group athletics [2]. Reported rates of SRC increased between the 2009/2010 and 2013/2014 academic years [3]; partially attributable to policy changes [4], improved accessibility to SRC educational resources [5–7], improved self-reporting behaviors [8], and the increased value of SRC assessments and clinical guidance [9].

The gold standard for concussion diagnosis continues to be the clinical examination and judgment of the medical provider [10–12]. As the landscape of SRC identification and management has rapidly changed over the previous two decades, there has also been an evolution of how the injury is characterized. Indeed, female student athletes have demonstrated to be more willing to disclose their concussions compared with male athletes when participating in sex-comparable sports [3]. Loss of consciousness (LOC) and post-traumatic amnesia (PTA) have been classically considered hallmarks of SRC, although estimates suggest less than 10% of concussed student athletes experience LOC [13, 14]. Further, neurodevelopmental conditions [15], mental health disorders (e.g., anxiety, depression), and familial and/or personal history of migraine headaches [16] may influence concussion risk [17–20]. Broadly, however, SRC literature lacks contemporary descriptive data of key demographic variables, medical history, and injury characteristics among collegiate student athletes as they relate to predictors of SRC.

The Concussion Assessment, Research, and Education (CARE) Consortium [21] offers a unique opportunity to characterize student athletes before and after sustaining a SRC and their injury characteristics. The purpose of the present study is two-fold: (1) to compare demographic, medical, and academic histories between a large cohort of concussed and non-concussed collegiate student athletes and (2) to evaluate differences in SRC injury characteristics based on biological sex and sport contact level. These findings may serve as a reference for clinicians and researchers to understand base rates of pre-morbid characteristics and identify those who may be more susceptible to injury within the collegiate setting. Further, these results may aid clinicians with SRC diagnosis, resulting in more precise and timely detection.

2 Methods

2.1 Study Design and Setting

Data were obtained from an ongoing prospective cohort study at 30 universities as a part of the CARE Consortium investigation. A detailed overview of the CARE Consortium methods have been described elsewhere [21]. Briefly, the CARE Consortium collects baseline and post-injury SRC-related data to assess the natural history of SRC among military service academy cadets and National Collegiate Athletic Association (NCAA) university student athletes [21]. Prior to study enrollment, participants provided written informed consent with their local institution and the US Military Human Research Protection Office [21]. This study was conducted in accordance with the Declaration of Helsinki.

Upon enrollment, all student athletes completed a series of baseline SRC assessments prior to the start of their athletic season or academic year to serve as a pre-injury baseline measure. In addition to SRC assessments, each participant completed the Demographics and Personal and Family Medical History unique case report form, which comprised several self-reported questions regarding demographic data (e.g. sex, race, ethnicity), sport and SRC histories, as well as personal and familial physical and mental health-related conditions [21].

All SRCs were diagnosed by the local medical staff using a common injury definition [22]. Detailed acute post-injury characteristics were collected by the local medical staff for student athletes who sustained a SRC, including details on the type of medical provider who made the initial diagnosis, the injury setting and venue (i.e., whether the SRC occurred at a home or away event), as well as whether the student

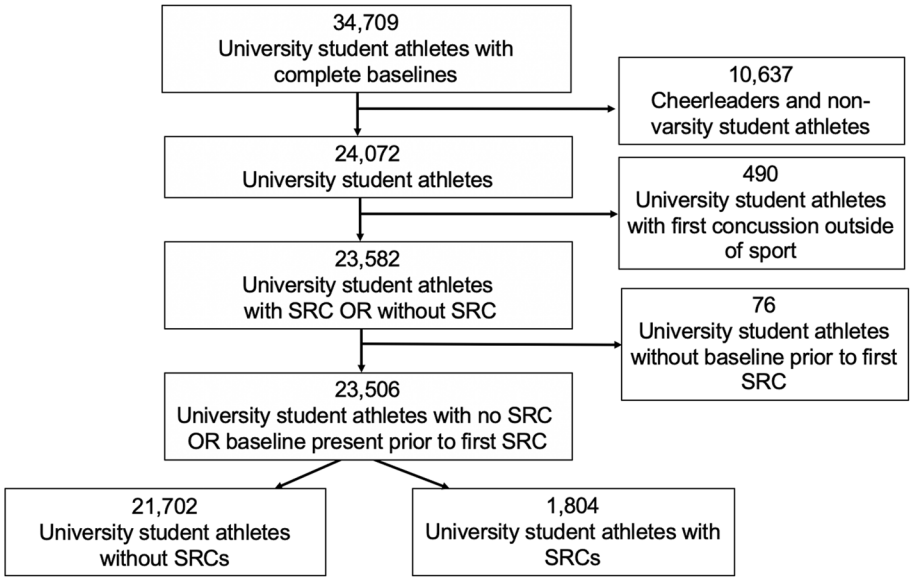

Fig. 1 Consort diagram of participating National Collegiate Athletic Association student athletes. Cheerleaders and non-varsity student athletes omitted from the analyses, as well as student athletes whose concussion occurred outside of sport or if they did not have a baseline assessment prior to their first sport-related concussion (SRC)

athlete experienced LOC, PTA, altered mental status, and/or motor impairment [21].

2.2 Participants

At the time of evaluation, the CARE dataset included 34,709 participants across 30 institutions with valid baseline assessments collected between August 2014 through May 2021. Post-SRC data were collected from August 2014 through May 2018. Participants were excluded from the analyses if they participated in a non-NCAA sanctioned sport (e.g., cheerleaders, non-varsity sport athletes, or service academy cadets who did not participate in university athletics) [$n = 10{,}637$], if their first concussion captured by the CARE Consortium occurred outside of sport ($n = 490$), or if they were missing a baseline assessment ($n = 76$) [23]. The final analyses included 23,506 student athletes, with 1804 diagnosed SRCs (Fig. 1). Sports ($n = 22$) were included in the present study if they accumulated ≥ 30 SRCs within the 7-year timeframe of data collection in order to preserve statistical power across these analyses. Frequencies of SRCs by sport, contact level, and sex are described in the Electronic Supplementary Material (ESM).

2.3 Statistical Methods

Descriptive statistics (frequencies and percentages, and measures of central tendency) were calculated for each demographic characteristic by SRC status (concussed vs non-concussed) among all student athletes, as well as stratified by sex and sport contact level (i.e., contact, limited contact, non-contact) [24]. Relationships between SRC and study characteristics were assessed using independent logistic regression models using odds ratios (ORs) and corresponding 95% confidence intervals (CIs) with SRC status as the outcome variable and each characteristic as the predictor variable. When outcomes have low prevalence (< 20%), ORs are a good approximation of risk ratios [25, 26]. Odds ratios were considered statistically significant if the 95% CI did not cross over 1.0 [23]. Next, we performed a set of stratified analyses examining the association between SRC and each characteristic within sex and contact-level categories. Finally, we conducted Pearson chi-square tests for parametric categorical variables and Kruskal–Wallis rank sum tests for nonparametric variables to assess the differences in SRC injury characteristics by sex and contact level. Data were analyzed using RStudio (Boston, MA, USA; version 4.0.4). Significance threshold was set a priori to $p < 0.05$.

3 Results

3.1 Demographics

A total of 1804 student athletes with SRC and 21,702 student athletes without SRC were included in this dataset. Table 1 provides the distribution of concussed and non-concussed student athletes across a number of demographic variables. When comparing sexes overall, female student athletes had lower odds of experiencing a SRC than male athletes (OR = 0.78; 95% CI 0.71, 0.86; Fig. 1 of the ESM). This relationship was not consistent across sport contact levels, with female athletes demonstrating higher odds of SRC for both limited and non-contact sports than male athletes (Table 2).

Table 1 Demographics

	All athletes		Male		Female		Contact		Limited contact		Non-contact	
	Non-concussed (N = 21,702)	Concussed (N = 1804)	Non-concussed (n = 12,600)	Concussed (n = 1154)	Non-concussed (n = 9099)	Concussed (n = 650)	Non-concussed (n = 10,334)	Concussed (n = 1407)	Non-concussed (n = 6943)	Concussed (n = 325)	Non-concussed (n = 4425)	Concussed (n = 72)
Sex, *n* (%)												
Male	12,600 (58.1%)	1154 (64.0%)					7555 (73.1%)	1027 (73.0%)	3432 (49.4%)	110 (33.8%)	1613 (36.5%)	17 (23.6%)
Female	9099 (41.9%)	650 (36.0%)					2779 (26.9%)	380 (27.0%)	3509 (50.6%)	215 (66.2%)	2811 (63.5%)	55 (76.4%)
Race, *n* (%)												
White	15,441 (73.5%)	1177 (67.1%)	8434 (69.5%)	663 (59.3%)	7006 (79.1%)	514 (80.8%)	6713 (67.3%)	869 (63.4%)	5114 (75.9%)	251 (79.7%)	3614 (84.5%)	57 (83.8%)
Black	3157 (15.0%)	387 (22.1%)	2353 (19.4%)	322 (28.8%)	804 (9.1%)	65 (10.2%)	2237 (22.4%)	355 (25.9%)	846 (12.6%)	31 (9.8%)	74 (1.7%)	1 (1.5%)
Multiracial	1428 (6.8%)	144 (8.2%)	835 (6.9%)	102 (9.1%)	593 (6.7%)	42 (6.6%)	719 (7.2%)	117 (8.5%)	498 (7.4%)	22 (7.0%)	211 (4.9%)	5 (7.4%)
Remaining races[a]	971 (4.6%)	46 (2.6%)	521 (4.3%)	31 (2.8%)	450 (5.1%)	15 (2.4%)	313 (3.1%)	30 (2.2%)	278 (4.1%)	11 (3.5%)	380 (8.9%)	5 (7.4%)
Ethnicity, *n* (%)												
Hispanic and/or Latinx	1744 (9.4%)	123 (8.0%)	1040 (9.9%)	74 (7.9%)	704 (8.6%)	49 (8.2%)	781 (9.2%)	76 (6.5%)	640 (10.3%)	39 (13.1%)	323 (8.2%)	8 (11.8%)
Non-Hispanic and/or Latinx	16,906 (90.6%)	1408 (92.0%)	9429 (90.1%)	861 (92.1%)	7476 (91.4%)	547 (91.8%)	7720 (90.8%)	1090 (93.5%)	5561 (89.7%)	258 (86.9%)	3625 (91.8%)	60 (88.2%)
NCAA Division, *n* (%)												
I	15,602 (71.9%)	1453 (80.5%)	8919 (70.8%)	942 (81.6%)	6680 (73.4%)	511 (78.6%)	7304 (70.7%)	1120 (79.6%)	4926 (70.9%)	269 (82.8%)	3372 (76.2%)	64 (88.9%)
II	2606 (12.0%)	180 (10.0%)	1547 (12.3%)	107 (9.3%)	1059 (11.6%)	73 (11.2%)	1372 (13.3%)	141 (10.0%)	895 (12.9%)	34 (10.5%)	339 (7.7%)	5 (6.9%)
III	3494 (16.1%)	171 (9.5%)	2134 (16.9%)	105 (9.1%)	1360 (14.9%)	66 (10.2%)	1658 (16.0%)	146 (10.4%)	1122 (16.2%)	22 (6.8%)	714 (16.1%)	3 (4.2%)

Values reported as count (frequency)

[a]Remaining races include Asian, American Indian/Alaskan Native, and/or Hawaiian/Pacific Islander

Furthermore, student athletes who self-identified as Black (OR = 1.61; 95% CI 1.42, 1.81) or multiracial (OR = 1.32; 95% CI 1.10, 1.59) demonstrated a greater odds of experiencing SRC than White-identifying student athletes (Fig. 1 of the ESM), whereas those who identified as "Remaining Races" (i.e., Asian, American Indian/Alaskan Native, and/or Hawaiian/Pacific Islander) demonstrated a lower odds of SRC (OR = 0.62; 95% CI 0.46, 0.84). Additionally, both Division II (OR = 0.74; 95% CI 0.63, 0.87) and Division III (OR = 0.53; 95% CI 0.45, 0.62) student athletes experienced reduced odds of SRC in comparison to Division I (Fig. 1 of the ESM).

3.2 Medical and Academic Histories

Table 3 provides the distribution of concussed and non-concussed student athletes across a number of medical and academic history variables. Odds ratios from the logistic regression indicated student athletes with SRC history (1 and 2+) demonstrated increasingly higher odds of sustaining another SRC overall (Fig. 2 of the ESM) and across both sexes (Fig. 2 of the ESM). Moreover, those who previously received academic assistance (OR = 1.35; 95% CI 1.11, 1.63), or who were diagnosed with a learning disorder (OR = 1.50; 95% CI 1.17, 1.93) or attention deficit hyperactive disorder (ADHD; OR = 1.38; 95% CI 1.16, 1.64) resulted in higher odds of sustaining a SRC (Fig. 2 of the ESM).

Additional associations are presented in Table 4 including increased SRC odds among male athletes (OR = 1.47; 95% CI 1.20, 1.81) and contact sport student athletes (OR = 1.40; 95% CI 1.16, 1.70) previously diagnosed with ADHD; however, this was not the case for female athletes. Conversely, female student athletes with a history of diagnosed depression (OR = 1.54; 95% CI 1.14, 2.07) presented with higher odds of sustaining a SRC than male athletes, in which where there was no association. In addition, both male (OR = 1.28, 95% CI 1.01, 1.64) and female (OR = 1.47; 95% CI 1.14, 1.89) student athletes with migraine headaches demonstrated higher odds of SRC (Table 4). Yet, only male student athletes with a history of non-migraine headaches were at increased odds of SRC (OR = 1.91; 95% CI 1.22, 2.98), in contrast to female student athletes, in which the relationship was not statistically significant (Table 4).

3.3 SRC Characteristics

Several noteworthy trends emerged regarding post-injury characteristics (Table 5). Most SRCs occurred during practice/training (66.1%), at the home site venue (78.8%) and were diagnosed by a physician (87.6%). Delayed symptom onset (DSO) has been previously defined [27] as symptom onset greater than 15 min after the injury occurred. As a result, DSO was reported in nearly a third of all SRCs (31.2%), which may have directly influenced the results of student athletes who were immediately removed from play (54.9%). In other words, there may have been a higher

Table 2 Odds ratios of SRC[a] based on demographics stratified by sex and sport contact level

Variable	Male	Female	Contact	Limited contact	Non-contact
Sex					
Male			Ref	Ref	Ref
Female			1.01 (0.89, 1.14)	1.91 (1.51, 2.42)*	1.86 (1.07, 3.21)*
Race					
Black	1.74 (1.51, 2.00)*	1.10 (0.84, 1.44)	1.23 (1.07, 1.40)*	0.75 (0.51, 1.09)	0.86 (0.12, 6.27)
Multiracial	1.55 (1.25, 1.94)*	0.96 (0.70, 1.34)*	1.26 (1.02, 1.55)*	0.90 (0.58, 1.40)	1.50 (0.60, 3.79)
White	Ref	Ref	Ref	Ref	Ref
Remaining races[b]	0.76 (0.52, 1.10)	0.45 (0.27, 0.77)*	0.74 (0.51, 1.08)	0.81 (0.44, 1.49)	0.83 (0.33, 2.09)
Ethnicity					
Hispanic and/or Latinx	0.78 (0.61, 1.00)	0.95 (0.70, 1.29)	0.69 (0.54, 0.88)*	1.31 (0.93, 1.86)	1.50 (0.71, 3.16)
Non-Hispanic and/or Latinx	Ref	Ref	Ref	Ref	Ref
NCAA Division					
I	Ref	Ref	Ref	Ref	Ref
II	0.66 (0.53, 0.81)*	0.90 (0.70, 1.16)	0.67 (0.56, 0.81)*	0.70 (0.48, 1.00)	0.78 (0.31, 1.94)
III	0.47 (0.38, 0.57)*	0.63 (0.49, 0.82)*	0.57 (0.48, 0.69)*	0.36 (0.23, 0.56)*	0.22 (0.07, 0.71)*

Asterisk (*) indicates statistical significance ($p < 0.05$)

NCAA National Collegiate Athletic Association, *Ref* reference

[a]Values reported as odds ratios (95% confidence intervals)

[b]Remaining races include Asian, American Indian/Alaskan Native, and/or Hawaiian/Pacific Islander

Table 3 Medical and academic history by sex and sport contact level

	All athletes		Male		Female		Contact		Limited contact		Non-contact	
	Non-concussed ($N=21{,}702$)	Concussed ($N=1804$)	Non-concussed ($n=12{,}600$)	Concussed ($n=1154$)	Non-concussed ($n=9099$)	Concussed ($n=650$)	Non-concussed ($n=10{,}334$)	Concussed ($n=1407$)	Non-concussed ($n=6943$)	Concussed ($n=325$)	Non-concussed ($n=4425$)	Concussed ($n=72$)
Previous diagnosed SRCs												
0	17,465 (80.5%)	1135 (62.9%)	10,034 (79.6%)	738 (64.0%)	7429 (81.6%)	397 (61.1%)	7660 (74.1%)	850 (60.4%)	5846 (84.2%)	234 (72.0%)	3959 (89.5%)	51 (70.8%)
1	3351 (15.4%)	497 (27.5%)	2067 (16.4%)	317 (27.5%)	1283 (14.1%)	180 (27.7%)	2118 (20.5%)	405 (28.8%)	851 (12.3%)	74 (22.8%)	382 (8.6%)	18 (25.0%)
2+	886 (4.1%)	172 (9.5%)	499 (4.0%)	99 (8.6%)	387 (4.3%)	73 (11.2%)	556 (5.4%)	152 (10.8%)	246 (3.5%)	17 (5.2%)	84 (1.9%)	3 (4.2%)
Academic assistance												
No	20,557 (94.7%)	1678 (93.0%)	11,884 (94.3%)	1061 (91.9%)	8670 (95.3%)	617 (94.9%)	9757 (94.4%)	1300 (92.4%)	6588 (94.9%)	309 (95.1%)	4212 (95.2%)	69 (95.8%)
Yes	1145 (5.3%)	126 (7.0%)	716 (5.7%)	93 (8.1%)	429 (4.7%)	33 (5.1%)	577 (5.6%)	107 (7.6%)	355 (5.1%)	16 (4.9%)	213 (4.8%)	3 (4.2%)
Learning disorder												
No	20,721 (97.2%)	1694 (95.9%)	12,021 (97.4%)	1078 (95.9%)	8699 (97.0%)	616 (96.0%)	9844 (97.2%)	1312 (95.6%)	6647 (97.2%)	310 (96.6%)	4230 (97.4%)	72 (100.0%)
Yes	587 (2.8%)	72 (4.1%)	321 (2.6%)	46 (4.1%)	266 (3.0%)	26 (4.0%)	284 (2.8%)	61 (4.4%)	191 (2.8%)	11 (3.4%)	112 (2.6%)	0 (0.0%)
Psychiatric disorder												
No	20,643 (97.1%)	1712 (97.1%)	12,132 (98.6%)	1102 (98.3%)	8510 (95.1%)	610 (95.0%)	9908 (98.2%)	1333 (97.3%)	6601 (96.7%)	309 (96.3%)	4134 (95.4%)	70 (97.2%)
Yes	608 (2.9%)	51 (2.9%)	174 (1.4%)	19 (1.7%)	434 (4.9%)	32 (5.0%)	186 (1.8%)	37 (2.7%)	222 (3.3%)	12 (3.7%)	200 (4.6%)	2 (2.8%)
ADHD												
No	19,698 (93.4%)	1601 (91.1%)	11,271 (92.9%)	1002 (89.9%)	8426 (94.1%)	599 (93.3%)	9231 (92.8%)	1232 (90.2%)	6394 (93.9%)	298 (93.4%)	4073 (94.0%)	71 (98.6%)
Yes	1394 (6.6%)	156 (8.9%)	865 (7.1%)	113 (10.1%)	529 (5.9%)	43 (6.7%)	716 (7.2%)	134 (9.8%)	416 (6.1%)	21 (6.6%)	262 (6.0%)	1 (1.4%)
Depression												
No	20,458 (96.6%)	1664 (95.7%)	12,018 (98.1%)	1080 (98.0%)	8439 (94.5%)	584 (91.8%)	9795 (97.8%)	1297 (96.4%)	6563 (96.0%)	301 (93.8%)	4100 (94.5%)	66 (91.7%)
Yes	726 (3.4%)	74 (4.3%)	237 (1.9%)	22 (2.0%)	489 (5.5%)	52 (8.2%)	217 (2.2%)	48 (3.6%)	272 (4.0%)	20 (6.2%)	237 (5.5%)	6 (8.3%)
Non-migraine headaches												
No	20,950 (98.4%)	1718 (97.6%)	12,191 (98.9%)	1097 (97.9%)	8758 (97.8%)	621 (97.0%)	9980 (98.7%)	1336 (97.7%)	6723 (98.5%)	312 (97.2%)	4247 (97.9%)	70 (97.2%)
Yes	332 (1.6%)	42 (2.4%)	134 (1.1%)	23 (2.1%)	198 (2.2%)	19 (3.0%)	135 (1.3%)	31 (2.3%)	105 (1.5%)	9 (2.8%)	92 (2.1%)	2 (2.8%)
Migraine												
No	19,861 (93.3%)	1609 (91.3%)	11,646 (94.5%)	1042 (93.0%)	8214 (91.7%)	567 (88.3%)	9493 (93.9%)	1250 (91.3%)	6360 (93.1%)	294 (91.6%)	4008 (92.5%)	65 (90.3%)
Yes	1420 (6.7%)	153 (8.7%)	679 (5.5%)	78 (7.0%)	741 (8.3%)	75 (11.7%)	622 (6.1%)	119 (8.7%)	471 (6.9%)	27 (8.4%)	327 (7.5%)	7 (9.7%)

Values reported as count (frequency)

ADHD attention deficit hyperactive disorder, *SRC* sport-related concussion

Table 4 Odds ratios of SRC[a] based on medical and academic histories stratified by sex and sport contact level

Variable	Male	Female	Contact	Limited contact	Non-contact
Previously diagnosed SRCs					
0	Ref	Ref	Ref	Ref	Ref
1	2.09 (1.81, 2.40)*	2.63 (2.18, 3.16)*	1.72 (1.52, 1.96)*	2.17 (1.66, 2.85)*	3.66 (2.12, 6.32)*
2+	2.70 (2.15, 3.39)*	3.53 (2.70, 4.62)*	2.46 (2.03, 2.99)*	1.73 (1.04, 2.87)*	2.77 (0.85, 9.06)*
Academic assistance					
No	Ref	Ref	Ref	Ref	Ref
Yes	1.45 (1.16, 1.82)*	1.08 (0.75, 1.55)	1.39 (1.12, 1.72)*	0.96 (0.57, 1.61)	0.86 (0.27, 2.75)
Learning disorder					
No	Ref	Ref	Ref	Ref	Ref
Yes	1.60 (1.17, 2.19)*	1.38 (0.92, 2.08)	1.61 (1.21, 2.14)*	1.23 (0.67, 2.29)	0.00 (0.00, Inf)
Psychiatric disorder					
No	Ref	Ref	Ref	Ref	Ref
Yes	1.20 (0.75, 1.94)	1.03 (0.71, 1.49)	1.48 (1.03, 2.11)*	1.15 (0.64, 2.09)	0.59 (0.14, 2.43)
ADHD					
No	Ref	Ref	Ref	Ref	Ref
Yes	1.47 (1.20, 1.81)*	1.14 (0.83, 1.58)	1.40 (1.16, 1.70)*	1.08 (0.69, 1.70)	0.22 (0.03, 1.58)
Depression					
No	Ref	Ref	Ref	Ref	Ref
Yes	1.03 (0.66, 1.61)	1.54 (1.14, 2.07)*	1.67 (1.22, 2.30)*	1.60 (1.00, 2.56)	1.57 (0.67, 3.66)
Non-migraine headaches					
No	Ref	Ref	Ref	Ref	Ref
Yes	1.91 (1.22, 2.98)*	1.35 (0.84, 2.18)	1.72 (1.16, 2.54)*	1.85 (0.93, 3.68)	1.32 (0.32, 5.46)
Migraine headaches					
No	Ref	Ref	Ref	Ref	Ref
Yes	1.28 (1.01, 1.64)*	1.47 (1.14, 1.89)*	1.45 (1.18, 1.78)*	1.24 (0.83, 1.86)	1.32 (0.60, 2.90)

Asterisk (*) indicates statistical significance ($p < 0.05$)

ADHD attention deficit hyperactive disorder, *Ref* reference, *SRC* sport-related concussion

proportion of student athletes who were immediately removed from play if symptoms were reported immediately following impact. Additional post-injury characteristics include altered mental status (49.6%), specifically LOC (4.7%), PTA (10.5%), and/or retrograde amnesia (RGA) (5.4%).

This investigation further unveiled differences in post-injury characteristics based on biological sex. The type of medical professional who confirmed the initial SRC diagnoses was associated with sex, with physicians diagnosing a greater proportion of SRCs in female athletes (91.2%) than male athletes (86.1%; $p = 0.013$). Additionally, a greater proportion of female student athletes experienced DSO (38.8%) than male athletes (26.8%; $p < 0.001$). However, more than half of male student athletes who sustained a SRC experienced an altered mental status (54.9%) in contrast to female athletes (40.4%; $p < 0.001$). Similar between-sex patterns were observed in LOC (male: 5.9%, female: 2.6%; $p = 0.002$), PTA (male: 13.6%, female: 5.1%; $p < 0.001$), and RGA (male: 6.8%, female: 2.8%; $p < 0.001$).

Differences in injury occurrence were reported across contact levels, with 80.6% of SRCs occurring during practice/training among non-contact sports, in contrast to 65.5% in contact sports and 65.8% in limited contact sports ($p = 0.030$). Increased incidence during practice/training may be related to greater exposure/opportunity for head impacts in contrast to competitions. There were also differences in DSO across contact levels, with increasing rates of delayed symptoms among non-contact and limited contact sports compared with contact sports ($p < 0.001$). Although no significant associations were noted between those who were removed from play and contact levels, total time before removal from play differed across each group, with decreasing time to removal with increasing contact level ($p = 0.005$). Furthermore, more contact sport student athletes experienced an altered mental status (52.7%) than limited contact (36.2%) and non-contact (48.6%; $p < 0.001$), with notable differences in PTA (contact: 11.6%, limited: 6.7%, non-contact: 5.6%; $p = 0.015$) and RGA (contact: 6.1%, limited: 2.6%, non-contact: 2.8%; $p = 0.025$). Additionally, frequency of post-injury LOC did not statistically differ across groups

Table 5 SRC characteristics

	SRC athletes ($N = 1804$)	Male ($n = 1154$)	Female ($n = 650$)	P-value[1]	Contact ($n = 1407$)	Limited contact ($n = 325$)	Non-contact ($n = 72$)	P- value[1]
Initial diagnosis				0.013				0.028
Physician[a]	397 (87.6%)	273 (86.1%)	124 (91.2%)		289 (85.8%)	88 (94.6%)	20 (87.0%)	
Specialist[b]	28 (6.2%)	26 (8.2%)	2 (1.5%)		27 (8.0%)	1 (1.1%)	0 (0.0%)	
Other[c]	28 (6.2%)	18 (5.7%)	10 (7.4%)		21 (6.2%)	4 (4.3%)	3 (13.0%)	
Injury setting				0.905				0.030
Competition	611 (33.9%)	392 (34.0%)	219 (33.7%)		486 (34.5%)	111 (34.2%)	14 (19.4%)	
Practice/training	1193 (66.1%)	762 (66.0%)	431 (66.3%)		921 (65.5%)	214 (65.8%)	58 (80.6%)	
Venue				0.239				0.613
Away	327 (18.6%)	196 (17.5%)	131 (20.7%)		250 (18.2%)	67 (21.3%)	10 (14.3%)	
Home	1383 (78.8%)	895 (79.8%)	488 (77.0%)		1085 (79.1%)	240 (76.4%)	58 (82.9%)	
Neutral site	46 (2.6%)	31 (2.8%)	15 (2.4%)		37 (2.7%)	7 (2.2%)	2 (2.9%)	
Delayed symptom onset				< 0.001				< 0.001
No	1201 (68.8%)	812 (73.2%)	389 (61.2%)		987 (72.3%)	179 (58.1%)	35 (49.3%)	
Yes	544 (31.2%)	297 (26.8%)	247 (38.8%)		379 (27.7%)	129 (41.9%)	36 (50.7%)	
Time to symptom onset (minutes)				0.375				0.169
Mean (SD)	324.0 (792.1)	311.2 (665.7)	338.5 (916.2)		328.9 (851.9)	307.0 (659.2)	335.6 (623.5)	
Median (Q1, Q3)	60.0 (15.0, 147.5)	45.0 (15.0, 120.0)	60.0 (20.0, 150.0)		45.0 (15.0, 120.0)	60.0 (30.0, 165.0)	60.0 (11.2, 285.0)	
Removed from play immediately				0.030				0.835
No	760 (45.1%)	466 (43.1%)	294 (48.6%)		590 (44.7%)	140 (46.4%)	30 (46.9%)	
Yes	926 (54.9%)	615 (56.9%)	311 (51.4%)		730 (55.3%)	162 (53.6%)	34 (53.1%)	
Time before removed from play, (minutes)				0.798				0.005
Mean (SD)	102.5 (539.6)	85.9 (364.1)	127.0 (725.5)		73.6 (322.2)	139.1 (560.0)	461.4 (1870.8)	
Median (Q1, Q3)	30.0 (10.0, 60.0)	30.0 (10.0, 60.0)	20.0 (10.0, 60.0)		20.0 (10.0, 60.0)	30.0 (15.0, 60.0)	40.0 (10.0, 90.0)	
Altered mental status				< 0.001				< 0.001
No	885 (50.4%)	504 (45.1%)	381 (59.6%)		649 (47.3%)	199 (63.8%)	37 (51.4%)	
Yes	871 (49.6%)	613 (54.9%)	258 (40.4%)		723 (52.7%)	113 (36.2%)	35 (48.6%)	
Loss of consciousness				0.002				0.512
No	1689 (95.3%)	1061 (94.1%)	628 (97.4%)		1318 (95.0%)	303 (96.5%)	68 (94.4%)	
Yes	84 (4.7%)	67 (5.9%)	17 (2.6%)		69 (5.0%)	11 (3.5%)	4 (5.6%)	
Duration of loss of consciousness (seconds)				0.837				0.767
Mean (SD)	14.2 (36.4)	13.8 (37.7)	16.6 (30.9)		13.7 (37.9)	10.3 (10.4)	34.5 (57.2)	
Median (Q1, Q3)	5.0 (2.8, 10.0)	5.0 (3.0, 10.0)	10.0 (2.0, 11.5)		5.0 (3.0, 10.0)	10.0 (2.5, 10.0)	8.5 (4.0, 39.0)	
Post-traumatic amnesia				< 0.001				0.015

Table 5 (continued)

	SRC athletes ($N=1804$)	Male ($n=1154$)	Female ($n=650$)	P-value[1]	Contact ($n=1407$)	Limited contact ($n=325$)	Non-contact ($n=72$)	P- value[1]
No	1582 (89.5%)	973 (86.4%)	609 (94.9%)		1223 (88.4%)	291 (93.3%)	68 (94.4%)	
Yes	186 (10.5%)	153 (13.6%)	33 (5.1%)		161 (11.6%)	21 (6.7%)	4 (5.6%)	
Duration of post-traumatic amnesia (minutes)				0.187				0.884
Mean (SD)	24.5 (48.3)	18.9 (33.0)	54.8 (91.3)		21.5 (41.7)	32.3 (49.1)	81.0 (146.2)	
Median (Q1, Q3)	5.0 (2.0, 20.0)	5.0 (2.0, 20.0)	10.0 (1.0, 60.0)		7.5 (2.0, 20.0)	3.0 (1.0, 60.0)	11.5 (2.5, 90.0)	
Retrograde amnesia				< 0.001				0.025
No	1675 (94.6%)	1050 (93.2%)	625 (97.2%)		1301 (93.9%)	304 (97.4%)	70 (97.2%)	
Yes	95 (5.4%)	77 (6.8%)	18 (2.8%)		85 (6.1%)	8 (2.6%)	2 (2.8%)	
Duration of retrograde amnesia (minutes)				0.528				0.331
Mean (SD)	28.4 (61.0)	24.0 (42.9)	46.8 (109.5)		23.0 (41.1)	82.8 (152.1)	1.0 (NA)	
Median (Q1, Q3)	5.0 (1.0, 27.5)	5.0 (1.0, 20.0)	12.5 (1.0, 30.0)		5.0 (1.0, 20.0)	22.5 (4.5, 52.5)	1.0 (1.0, 1.0)	
Motor impairment				0.191				0.003
No	782 (82.3%)	484 (81.1%)	298 (84.4%)		581 (81.0%)	172 (89.6%)	29 (70.7%)	
Yes	168 (17.7%)	113 (18.9%)	55 (15.6%)		136 (19.0%)	20 (10.4%)	12 (29.3%)	
Taken to hospital				0.345				0.357
No	1689 (95.5%)	1080 (95.8%)	609 (94.9%)		1329 (95.8%)	294 (94.5%)	66 (93.0%)	
Yes	80 (4.5%)	47 (4.2%)	33 (5.1%)		58 (4.2%)	17 (5.5%)	5 (7.0%)	
Imaging completed				0.115				0.476
CT	38 (80.9%)	22 (48.9%)	16 (50.0%)		25 (45.5%)	11 (64.7%)	2 (40.0%)	
MRI	9 (19.1%)	8 (17.8%)	1 (3.1%)		8 (14.5%)	1 (5.9%)	0 (0.0%)	
None	30 (39.0%)	15 (33.3%)	15 (46.9%)		22 (40.0%)	5 (29.4%)	3 (60.0%)	
Positive imaging findings				0.511				> 0.99
No	41 (95.3%)	24 (92.3%)	17 (100.0%)		28 (93.3%)	11 (100.0%)	2 (100.0%)	
Yes	2 (7.7%)	0 (0.0%)	2 (4.7%)		2 (6.7%)	0 (0.0%)	0 (0.0%)	

Differences between groups were calculated using the Pearson's chi-squared test for categorical variables and the Kruskal–Wallis rank sum test for nonparametric continuous variables

CT computed tomography, *MRI* magnetic resonance imaging, *Q* quartile, *SD* standard deviation, *SRC* sport-related concussion

[a]Doctor of osteopathic medicine, primary care physician, team physician

[b]Neurologist, neurosurgeon, orthopedic physician

[c]Neuropsychologist, physical therapist, physician assistant, other

(contact: 5.0%, limited: 3.5%, non-contact: 5.6%; $p=0.512$). Last, a greater frequency of non-contact student athletes experienced motor impairments (29.3%) in comparison to contact (19.0%) and limited contact student athletes (10.4%; $p=0.003$).

4 Discussion

The first objective of this investigation was to compare premorbid demographic, medical, and academic factors among concussed and non-concussed collegiate student athletes.

Several intrinsic (biological sex, race, concussion history) and extrinsic factors (Division level, participation setting) were found to influence SRC risk. When considering all student athletes participating across all sports, female athletes demonstrated lower odds of sustaining a SRC than male athletes, whereas the opposite was noted when stratified by contact level. These results suggest male student athletes participating in contact sports incurred a substantially higher number and percentage of SRCs relative to the female contact-sport category, while this trend was reversed for limited and non-contact sports. However, it is critical to note there are several potential contributing factors regarding these relationships that may provide additional context to these findings. Specifically, there was a disproportionate number of male student athletes who participated in contact sports relative to female athletes (Tale 1 of the ESM). Although previous work suggests female athletes sustain a higher risk of concussion relative to men in sex-comparable sports [3, 28], the quantity of male student athletes participating in contact sports and the amount of contact-related SRCs (in particular during football) skews this finding. While preceding work suggests there are several sex-related differences in SRC incidence, risk, and severity [28–30], future research should investigate how extrinsic (e.g., mechanism of injury, setting) and/or intrinsic factors, such as sex hormones (e.g., progesterone, estrogen, testosterone) [31] and social norms (e.g., differences in help-seeking behavior), contribute to differences in SRC across the gender spectrum.

Further, student athletes who identified as Black or multiracial demonstrated increasingly higher odds of SRC relative to those who identified as White. These findings are noteworthy as they contrast prior work, suggesting those who identify as White are more likely to identify the signs and symptoms of SRC [32] and report them to a healthcare provider than those who identify as Black [33]. However, data from the NCAA [34] suggest 40% of student athletes who participated in the football season self-identify as Black, which accounted for a third of all concussions included in this study. Collectively, these findings do not suggest a causal relationship between race and SRC incidence, but rather highlight sport participation by race and subsequent injuries. Future research should consider how race and other social constructs influence SRC risk within collegiate athletics.

Injury risk was also found to be influenced by setting, including NCAA Division level and site. For instance, student athletes participating in NCAA Division I athletics demonstrated higher odds of SRC than Divisions II and III. A majority of SRCs in the current study occurred at the Division I level (72%), which may be driving these results. Division I schools made up the majority of CARE sites (> 50%), and those schools likely have greater access to medical providers to disclose an injury than those participating in Division II or III [35], resulting in greater SRC reporting. Indeed, previous research concluded Division II student athletes were more inclined to disclose their SRC symptoms to a coach than an athletic trainer in contrast to student athletes participating in Division I or III athletics [36]. In addition, a greater proportion of SRCs occurred during practices (66%) than competitions (33%). This finding is consistent with an evaluation of football-related SRCs [37] and NCAA data from 2014/15 to 2018/19 academic years [38] reporting similar concussion frequencies between practices (49%) and competition (51%). Notably, concussion rates during competition (10.39 per 10,000 athlete exposures) are approximately fourfold that of practices (2.52 per 10,000 athlete exposures) [38], yet there are substantially more practices than competitions during an athletic season.

Student athletes with a history of one or more diagnosed SRCs were at a greater likelihood of sustaining a subsequent SRC. This notable increase in the likelihood of SRC complements foundational work by Guskiewicz and colleagues, suggesting a cumulative effect or "dose–response" with each successive SRC among a cohort of collegiate American football players [39]. Specifically, researchers concluded those with a history of one (1.4; 95% CI 1.0, 2.1), two (2.5; 95% CI 1.5, 4.1), or three or more SRCs (3.0; 95% CI 1.6, 5.6) were more likely to sustain another SRC in contrast to those without prior SRCs [39, 40]. The present study includes student athletes across 22 NCAA sports, which highlights similar trends in SRC risk based on injury history. Moreover, student athletes with pre-morbid learning disabilities or ADHD demonstrated higher odds of sustaining SRC than those without, which is in parallel with Alosco et al. [20] and Iverson et al. [41] who suggested ADHD may be a significant risk factor for SRC. Largely, these findings highlight several potential risk factors may influence the likelihood of sustaining SRCs. Clinicians should consider incorporating prevention strategies to reduce the likelihood of SRCs among high-risk student athletes and consider the presence of these factors in their clinical evaluation. Furthermore, clinicians may consider these findings when providing SRC educational materials to sports teams at the beginning of each season to further encourage disclosure behaviors, especially among student athletes and/or teams who may be at a higher risk of sustaining SRCs (e.g., contact sports).

This study further sought to evaluate differences in SRC post-injury characteristics. Across sexes, a greater frequency of SRCs occurred in male (64%) than female athletes (36%). Notably, more female athletes reported DSO than male student athletes; however, more male athletes experienced an altered mental status including LOC, PTA, and RGA in contrast to their female counterparts. These findings complement prior research concluding male athletes reported more amnesia-type symptoms [42] and LOC [43] than their female equivalents. This suggests male student athletes may

be less likely to report milder injuries (i.e., SRCs that do not result in an altered mental status). Indeed, researchers suggest stakeholder attitudes and situational context influence SRC reporting behaviors [44]. When assessing motivations for SRC nondisclosure, a majority of student athletes indicated they believed a blow to the head was not serious enough to be reported [45], even though this may result in a SRC diagnosis. This may, in part, be due to a lack of education regarding the common signs and symptoms of SRC. For instance, previous work estimated < 10% of SRCs resulted in LOC [13]; yet LOC continues to be mistakenly considered as one of the most significant signs of SRC despite its relatively low incidence [46, 47]. In addition, McCrea and colleagues [14] reported 8% of SRCs among a cohort of high school and collegiate American football players experienced PTA. However, these studies do not capture the evolution of SRC clinical management in recent years [10, 12], which may ultimately influence perceptions of SRC characteristics and reporting behaviors. In sum, the present study describes current post-injury characteristics stratified by biological sex and sport contact level, which may be more generalizable to the current sports participation and injury risk trends within the NCAA. Clinicians may further utilize these findings to educate student athletes on the seriousness of SRC in order to promote early reporting behaviors, especially among student athletes who may not be as likely to report their injury to a medical provider.

Despite the breadth of these findings, there are several limitations worth noting. First, data in these analyses do not include military service academy cadets who did not participate in university-level sports, and therefore may not be applicable to SRCs sustained in all military academy settings (e.g., combative training). In order to preserve statistical power, sports that accumulated < 30 SRCs during the data collection period (field hockey, beach volleyball, fencing, field event [track], golf, rifle, rowing/crew, tennis; Table 1 of the ESM) were omitted from the analyses, reducing the generalizability of these findings to all NCAA-sanctioned sports. Further, 34% of SRCs were attributed from American football, which only includes male student athletes. Therefore, grouping football within the contact sports category may be skewing these findings. Another limitation worth noting is the non-uniform racial proportions between sports, which was unaccounted for in the analyses. Future studies may consider stratifying these statistical analyses by race to compare concussion risk and injury characteristics. In addition, we were unable to account for differences across ethnicities because of the low sample of Hispanic and/or Latinx student athletes collected in this sample. Additionally, institutions who agree to participate in the CARE Consortium are provided resources to capture SRC-related data, which may not be representative of all SRCs sustained within the NCAA. Further, there was no control for athlete exposures, which may place certain student athletes at a greater risk for SRC than others. Additionally, the data collected from the demographics questionnaires were self-reported, which is a common limitation among survey-based research. Last, the relevance of these findings is limited to NCAA athletics. Additional research is warranted to investigate trends in SRC risk and post-injury characteristics among other universities, such as those who participate in the National Association of Intercollegiate Athletics and/or junior colleges within the National Junior College Athletic Association.

5 Conclusions

The present study highlights the complex heterogeneity of SRCs resulting in several individualized clinical presentations. Although altered mental status was once considered to be a classic characteristic of SRC, LOC (4.7%), RGA (5.4%), and PTA (10.5%) were infrequently reported among the current sample of collegiate student athletes who sustained SRCs, all of which supports contemporary definitions of concussion (e.g., Concussion in Sport Group) [10]. These differences were further noted across sexes, with male athletes demonstrating more incident LOR, PTA, and RGA. Similar trends were noted among contact level, with contact sport athletes experiencing an altered mental status more often than limited and non-contact sport athletes. Last, female athletes were more likely to experience DSO, which may negatively influence overall recovery trajectories, as they were likely not immediately removed from play. Overall, these findings provide new perspectives of SRC risk factors and post-injury characteristics using a large cohort of collegiate student athletes.

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Ethics Approval Participants provided written informed consent prior to study participation. The Human Research Protection Office approved the study and each participating site's institutional review board approved this study. All study operations were in accordance with the standards outlined in the Declaration of Helsinki.

Consent to Participate All participants provided written informed consent prior to participation.

Availability of Data and Material Data may be available upon reasonable request.

Code Availability Not applicable.

Authors' Contributions AKM was involved in the study conception and design, as well as the interpretation of data, manuscript drafting, and final approval. KMM and SMP were involved in the study conception and design, statistical analyses, and interpretation, in addition to reviewing subsequent drafts of the manuscript and providing final approval of the manuscript. All remaining authors were responsible for the study conception, design, and providing generous feedback on manuscript drafts. All authors read and approved the final manuscript.

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